MIT SCHOOL OF COMPUTING
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A Leap Towards World Class Education

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Subject: Programming Language Roll No.: 01 Exp. No.:
Name of the Experiment: Assignment -02

Marks	Teacher's Signature with date
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Performed on:
Submitted on:

1. List the python data type and state one Key characteristic of each with respect to mutability or ordering ?

→ i) List :- Ordered and mutable (elements can be changed).
ii) Tuple :- Ordered ~~and~~ and immutable (cannot be modified).
iii) String :- Ordered and immutable sequence of characters.
iv) Set :- Unordered and mutable, store unique elements only.
v) Dictionary :- Ordered (Python 3.7+), mutable, stores data as key value pairs.

2. Explain how indexing and slicing works in python sequences. Illustrate how they differ when applied ~~to~~ to string and list ?

→ ~~Indexing~~ : It is used to access a part of the sequence
Slicing : using [start : end : step].
Indexing :- It is ~~is~~ used to access a ~~part of the sequence~~ ~~the~~ single element using its position (start from 0).

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

Example :-

s = "Python"

I = [10, 20, 30, 40]

Indexing :

s[1:4] → 'yth'

I[1:3] → [20, 30]

Difference :

→ string cannot be modified after slicing (immutable).

→ List can be modified after slicing (mutable).

3. Write a python program that:
a) Uses of indexing and slicing on a string, list, and
b) Applies atleast two built in function to apply display meaningful results.

→ text = "Aaryan"
print ("first character : ", text [0])
print (" Substring : ", text [0 : 3]
print (" length of string : ", len (text)).
print (" Uppercase : ", text.upper ())

Output :
First character : ~~f~~ A
Substring : Aar
length of string : 7
Uppercase : AARYAN

4. Analyze the difference betw equality (==) and identity (is) when used ~~w~~ with sequence type. Discuss how object mutability and memory reference affect program behaviour.

→ Equality (==) checks whether values are same.
Identity (is) checks whether bothe variable print to the same ~~a~~ memory location.

Example -
a = [1, 2, 3]
b = [1, 2, 3]
print (a == b)
print (a is b)
Output : True

~~Explan~~ Explanation:
• lists are mutable and stored in different memory location.
• (==) compreess content.
• (is) compares memory reference.

5. Evaluate the suitability of lists, sets, and dictionaries for string and managing unique student rewards. Justify which data structure is must efficent and why?

→ List : Allows duplicates, slower searching.
Set : Stores only unique value, no indexing.
Dictionaries : stores unique student IDs as Keys and detail as value.
Most efficient - Dictionary, because it allow fast access, unique keys and structured data storage.

6. Design and implement a python program that uses:
- Appropriate sequence type
- Indexing or slicing
- Built-in function

to manage and process student data efficiently.

→

```
students = { 101 : "Amit", 102 : "Neha", 103 : "Rahul" }
print ("students 101 :", students [101])
print ("Total students :", len (students))
print ("Student IDs :", list (students.Key ()))
```

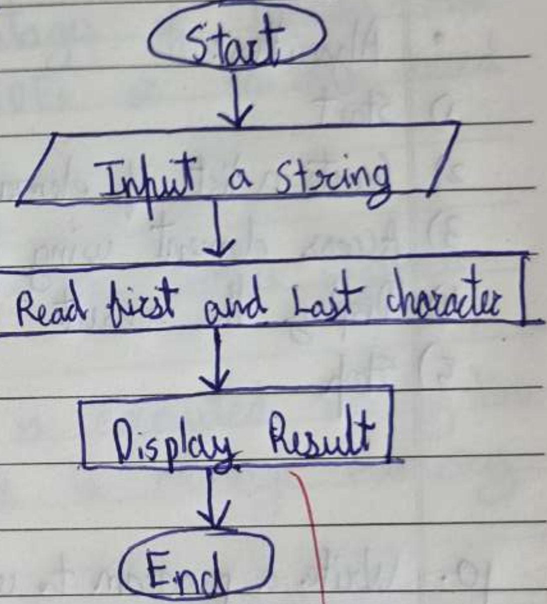
Output :

```
Student 101 : Amit
Total students : 3
Student IDs : [101, 102, 103]
```

7. Write a program to print the first and last character of a string?

→.
```
s = input ("Enter a string : ")
print ("First character : ", s [0])
print ("Last character : ", s [-1])
```

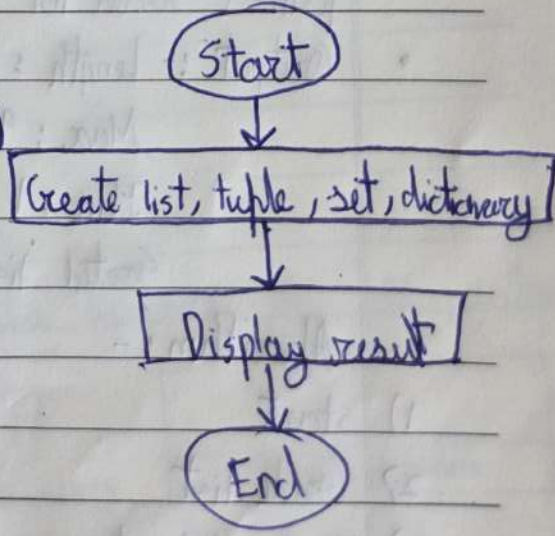
• Output : First character : P
  last character : n



• Algorithm :
1) start
2) Read a string
3) Find the first character
4) Find the last character
5) Display result
6) stop

8. Write a program to create a list, tuple, set & dictionary?

→
```
list = [1,2,3]
tuple = (4,5,6)
string = {7,8,9}
dictionary = {1:"one", 2:"Two"}
print (list, tuple, string, dictionary)
```



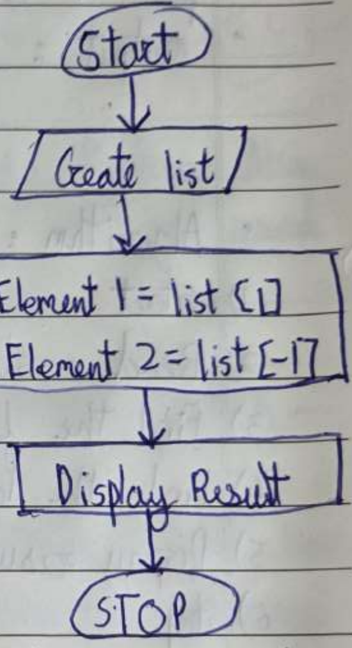
• Algorithm:
1) start
2) Create a list, tuple, set, dictionary
3) Display result
4) Stop

• Output : [1,2,3]
  (4,5,6)

9. Write a program to understand the concept of indexing.

→ • array = [10, 20, 30, 40]
print ("Element at index 1: " array [1])
print ("Element at index -1: " array [-1])

• Output : Element at index 1 : 20
Element at index -1 : 40

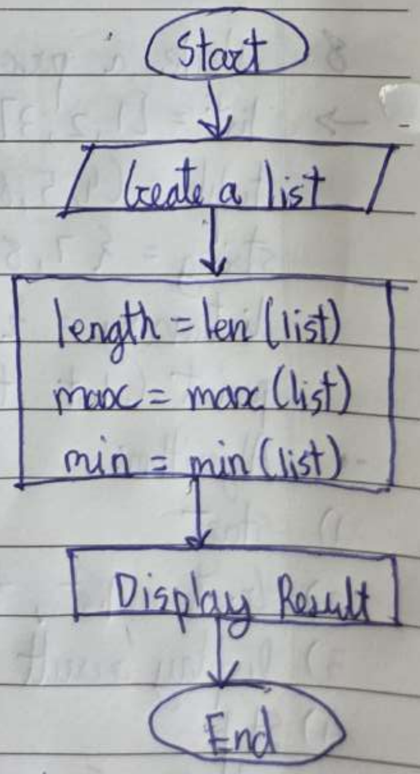


• Algorithm:
1) Start
2) Create a list of elements
3) Access element using positive & negative index
4) Display the result
5) Stop

10. Write a program to understand the concept of built-in functions?

→ • num = [5, 2, 9, 1]
print ("Length: ", len(num))
print ("Max : ", max (num))
print ("Min : ", min (num))
print ("Sorted list : ", sorted (num))

• Output :- Length : 4
Max : 9
Min : 1
Sorted list: [1, 2, 5, 9]



• Algorithm:-
1) Start
2) Create list
3) Use len () function, max () function & min () function
4) End